MILITARY SPECIFICATION

# TROLLEYS, I-BEAMS, PLAIN AND GEARED ADJUSTABLE AND NONADJUSTABLE 250- TO 8000-POUND CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of I-beam trolleys includ five sizes of plain trolleys and one size of geared trolley. These six trol include four adjustable and two nonadjustable trolleys.

1.2 Classification. The trolleys shall be of the following types and siz as specified (see 6.2):

Type I Plain

250 pound, nonadjustable
1,000 pound, adjustable
2,000 pound, adjustable
4,000 pound, adjustable
8,000 pound, adjustable

Type II Geared

2,000 pound, nonadjustable

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DoDISS) specifi in the solicitation, form a part of this specification to the extent specifi herein.

Beneficial comments (recommendations, additions, deletions) and any pertinen data which may be of use in improving this document should be addressed to: Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3950

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SPECIFICATION

FEDERAL

| | |
|---|---|
| PPP-B-601 | - Boxes, Wood, Cleated Plywood. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage. |

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Each trolley shall be in accordance with table 1 and shall consist of two frames, each equipped with one or two flanged trolley wheel(s) and axle(s) and bumpers as specified herein. The 250 pound size trolley shall have one flanged trolley wheel and axle on each side frame. The bumpers shall protect the wheels and also shall act as safety lugs to hold the load in case of



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wheel or axle failure. The side frames shall be connected at the bottom by a heavy steel equalizer pin on which a steel hook plate is centrally located to assure equal distribution of the load on the trolley wheels. The distance between the side frames shall be adjustable on adjustable trolleys to fit the beams as specified in table 1. The bottom hole in the suspension plate shall be not less than the diameter shown in table 1.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the trolleys may be newly fabricated from recovered materials to the maximum extent practicable, provided the trolleys produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the trolleys.

3.4 Characteristics. The trolleys shall conform to table 1. All load-sustaining parts shall incorporate a safety factor of not less than five based on ultimate strength of the material. There shall be no permanent distortion as the results of applying a load equal to three times the load-carrying capacity as specified in table 1.

TABLE I. Trolley requirements.

| Size and load carrying capacity (pounds) | Type | Adjustable or non-adjustable | I-Beam trolley rail size | Minimum wheel diameter tread (inches) | Suspension plate hole diameter (inches) |
|---|---|---|---|---|---|
| 250 | I, Plain | Nonadjustable | 2-inch-wide flange | 1-1/4 | 1-1/8 |
| 1,000 | I, Plain | Adjustable | 6 inches high, 12.5 pounds to 7 inches high, 15.3 pounds | 2 | 1-1/8 |
| 2,000 | I, Plain | Adjustable | 6 inches high, 12.5 pounds to 8 inches high, 18.4 pounds | 2-3/4 | 1-3/16 |



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TABLE I. Trolley requirements. (Continued)

| Size and load carrying capacity (pounds) | Type | Adjustable or non-adjustable | I-Beam trolley rail size | Minimum wheel diameter tread (inches) | Suspension plate hole diameter (inches) |
|---|---|---|---|---|---|
| 4,000 | I, Plain | Adjustable | 6 inches high, 12.5 pounds to 15 inches high, 50 pounds | 3-3/8 | 1-9/16 |
| 8,000 | I, Plain | Adjustable | 15 inches high, 50 pounds to 24 inches high, 79.9 pounds | 6 | 2 |
| 2,000 | II, Geared | Nonadjustable | 6 inches high, 12.5 pounds to 10 inches high, 25.4 pounds | 2-3/4 | 1-3/16 |

3.5 Side frames. Side frames shall be of rolled steel plate. Side frames shall mount the trolley wheels, bumpers, safety lugs and trolly blocks. Trolley blocks shall be riveted or welded to the side frames.

3.6 Trolley wheels. Unless otherwise specified herein, all trolley wheels shall be forged, machined, or formed from steel. When specified (see 6.2), trolley wheels shall be cast iron with chill treads or nodular iron wheels. The wheel treads shall be crowned so as to be suitable for operation on both sloped-flanged and flat-flanged I-beams.

3.6.1 Trolley-wheel axles. The axles for the trolley wheels shall be of steel.

3.6.2 Trolley wheel bearings. The trolley wheels shall each be equipped with a sealed tapered roller bearing.

3.7 Type II geared trolleys. The type II, geared trolley shall be propelled along the supporting beam by not less than two geared trolley wheels. These geared trolley wheels shall be driven by a pinion gear mounted on the end of a shaft, extending through the side frame and rotated by a hand-chain wheel. The hand-chain shall be of link type, shall be continuous, and shall extend not less than 8 feet from the center of the hand-chain wheel. Guards shall extend part of the way around the hand-chain wheel to prevent the hand-chain from coming out of the pockets in the wheel. All gears shall have cut teeth.

3.7.1 Hand-chain wheel, geared trolley wheels, and pinion gear. The hand-chain wheel shall be of steel or cast iron. The geared trolley wheels and pinion gear shall be of steel.



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3.7.2 Shaft. The hand-chain wheel shaft shall be of steel.

3.7.3 Bearings. The bearings supporting the hand-chain wheel shaft shall be straight roller bearings or self lubricating bushings.

3.8 Identification marking. Each trolley shall be identified with an identification plate containing the following information: Ownership designation, nomenclature, model, contract number, trolley serial number, capacity, national stock number (NSN), and date manufactured. The plate shall conform to MIL-P-514, type III, composition C, (GG-S-455) type I, grade A, class D unless otherwise specified. The identifying data shall be stamped in letters in accordance with MIL-P-514. The plate shall be securely attached to the trolley with bolts, rivets or screws and shall be furnished and mounted by the contractor.

3.9 Treatment and painting. The parts of the trolley normally painted shall be cleaned, treated and painted in accordance with the contractor's commercial practice.

3.10 Workmanship. All parts, components assemblies of the trolley including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux and other harmful extraneous materials. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required. No parts shall be damaged or impaired in any way, and parts shall show no evidence of rust or corrosion.

4. QUALITY ASSSURANCE PROVISIONS

4.1 Responsiblility for inspection. Unless otherwise specified in the contract, the contractor is responsible for all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to requirements prescribed.

4.2 Classification of inspections, evaluation and demonstrations. Inspections, evaluations, testing, and demonstrations shall be classified as follows:

a. First article evaluation, test and demonstration (see 4.3).
b. Inspection of packaging (see 4.5).

4.3 First article evaluation, test and demonstration test report. Prior to delivery the manufacturer shall make one or more trolleys available, selected at random from trolleys being produced from production tooling for first article evaluation and test. The trolley and its components shall be compared to this document, the manufacturers calculations and test data, to verify compliance with this specification. The trolley shall then be subjected to test and



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demonstration or test as specified herein. Permanent deformation, malfunction, failure to meet performance requirements, failure to complete any demonstration or test shall be cause for rejection of the trolley. If such deficiencies do arise, they shall be corrected and affected demonstration or test shall be repeated as required. The contractor shall prepare and submit the first article demonstration and test report to the contracting officer for approval within the time frame specified (see 6.2). The report shall include a detailed description of the examination, demonstration operational performance, deficiencies encountered and actions taken to correct the deficiencies.

4.3.1 First article (preproduction) examination.

4.3.1.1 Pretest examination. Prior to testing, one trolley shall be examined for compliance with 4.3.1.2. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Examination. The trolley shall be examined for the following defects:

101. Dimensions not as specified.
102. Grease seals do not retain lubricants.
103. Safety lugs not as specified.
104. Misalignment of axles.
105. Workmanship not as specified.
106. Materials not as specified.
107. Identification marking missing.
108. Treatment and painting not as specified.
109. Repairing or maintenance tools missing.
110. Welding not as specified.
111. Machined surfaces not as specified.
112. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the trolley.

4.4 Tests.

4.4.1 Safety factor. Mount the trolley on the largest I-beam as specified in table I for the size of trolley being tested. Hang a load on the trolley equal to not less than two times the load-carrying capacity as specified in table I. Examine the trolley for any evidence of distortion caused by the load placed upon the trolley. Evidence of any permanent distortion of the trolley shall constitute failure of this test.

4.4.2 Operation. Install the trolley on the smallest I-beam as specified in table I for the size of trolley being tested. Load the trolley with not less than the rated load and move the trolley back and forth along the beam for not less than 20 feet. Repeat this cycle not less than 10 times. Install the trolley on the largest I-beam specified in table I for the size of trolley being tested and operate the trolley in the same manner and for the same number of cycles as with the smallest size beam. Any breakage, permanent deformation, or operating difficulty such as sticking, binding, or jamming shall constitute failure of this test.



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4.5 Inspection of packaging. Packed trolleys shall be examined for the following major defects:

113. Preservation not applied to trolleys if required for level A.
114. Trolleys not individually unit packed as specified for level A and commercial.
115. Shipping container for level A not as specified.
116. Blocking and cushioning not used as specified.
117. Trolleys not consolidated as specified for commercial.
118. Marking not as specified.

Any packaging defects shall be corrected and reinspection performed before the affected trolleys shall be considered as suitable for shipment.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Unpainted, ferrous metal surfaces of each trolley shall be coated with type P-1 preservative as described in MIL-P-116. Each trolley, except for the 8,000 pound size, shall be individually unit packed in a box conforming to PPP-B-636, type CF or SF, class weather-resistant, grade as appropriate for the weight of the trolley. Each trolley shall be braced and cushioned within its box in a manner to prevent movement of the trolley and damage to the box. Boxes shall be closed in accordance with the box specification.

5.1.2 Commercial. Each trolley shall be individually unit packed in containers suitable for the weight involved and which will provide protection to the trolleys against deterioration and damage. Preservatives and cushioning shall be used as deemed necessary to protect critical surfaces from corrosion and fragile areas from damage, respectively.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Trolleys, unit packed as specified in 5.1, shall be packed in boxes conforming to PPP-B-601, overseas type, style I. Gross weight of shipping containers shall not exceed 200 pounds. Blocking shall be used as necessary to prevent movement of trolley(s) within shipping containers. Boxes shall be closed and reinforced in accordance with the box specification. Strapping shall be zinc coated flat steel.

5.2.2 Commercial. Unit packed trolleys shall be consolidated in shipping containers or on pallets in a manner which will assure safe and undamaged delivery to the destination and which will be acceptable to carriers.

5.3 Marking. Marking of unit packs and shipping containers for level A shall be in accordance with MIL-STD-129. Marking for commercial packaging, both unit packs and shipping containers or pallets shall be in accordance with ASTM D 3951 In addition, shipping containers or pallets shall show the weight and cube of the pack.



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## 6. NOTES

6.1 Intended use. The trolley is intended for use with an I-beam trolley rail, supported overhead, and a hoist to lift and transport loads from place to place along the line of the overhead trolley rail.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2).
c. When a first article test is not required (see 3.2).
d. When cast iron or nodular iron trolley wheels are required (see 3.6).
e. Degree of preservation and of packaging (see 5.1, 5.2).

6.3 First article. When a first article inspection is required, the item should be a first produced model. The first article should consist of one unit of each type and size trolley procured. The contracting officer should include specific instructions in procurement documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article(s).

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts list and suppliers maintenance and operation manual to be furnished with each trolley.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of trolleys.

Custodian:
Army - ME
Air Force - 99

User activity:
Army - SM

Preparing activity:
Army - ME

Project 3950-0243



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-52306C | Trolleys, I-Beams, Plain and Geared Adjustable and Nonadjustable 250- to 8000-Pound Capacity |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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